



**Lab Project**  
**Network Security and Cryptography Lab**  
**Course Code: CIS 311L**

| Submitted by                                                                                              | Submitted to                                                                             |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
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## **Introduction**

Cryptography is an important part of network security that help protect confidential information from unauthorized access and modification. This task focuses on practical applications of different cryptographic techniques, including AES-128 encryption, SHA-256 hashing, substitution-cipher decryption, and PGP encryption. AES-128 encryption and decryption were implemented using JavaScript, while SHA-256 hashing was implemented using Python to verify message integrity and demonstrate the avalanche effect. A Python-based substitution-cipher decryption program was also developed to recover plaintext from ciphertext. Finally, PGP tools were used to demonstrate secure file encryption and decryption using public and private keys. The main purpose of this lab is to understand how different cryptographic techniques can be applied to protect data, verify data integrity, recover encrypted information, and securely communicate confidential files.

## 1. AES-128 Encryption:

AES (Advanced Encryption Standard) is a secure symmetric-key encryption algorithm. In this task, a JavaScript application was developed to encrypt and decrypt a user message using AES-128. The same secret key was used for both encryption and decryption. After encryption, the message became unreadable (ciphertext). When the same key was used for decryption, the original message (plaintext) was successfully restored.

**Figure 1: AES Encryption Output:**

← ↻ 127.0.0.1:5500/index.html ☆

### AES Encryption & Decryption

Tomas Shelbi- SIGMA

Encrypt Decrypt

#### Encrypted Text

U2FsdGVkX1/BkdxFFaCZA2N10B2SHnCF3Qg6nnOKKwfw5kACKVOG5bijSOCRHTjO

#### Decrypted Text

Tomas Shelbi- SIGMA

## 2.2 SHA-256 Hashing:

SHA-256 is a cryptographic hash function that generates a fixed-length hash value. A Python program was used to calculate the hash of a message. After changing one character of the original message, a completely different hash value was produced. This demonstrates the avalanche effect, where a small change in input causes a significant change in output.

**Figure 2: SHA-256 Output:**

```
8 print("Original Message :", message)
9 print("SHA-256 Hash      :", hash1)
10
11 print("\n-----\n")
12
13 modified_message = "Confidential Project Datb"
14
15 hash2 = hashlib.sha256(modified_message.encode()).hexdigest()
16
17 print("Modified Message :", modified_message)
18 print("SHA-256 Hash      :", hash2)
19
20 print("\n-----\n")
```

Problems Output Debug Console **Terminal** Ports

Hashes Do Not Match  
Avalanche Effect Demonstrated  
PS C:\Users\kaysu\NSC LAB>

File (NSC LAB) Ln 3, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.14.7

### 3.1 Substitution Cipher:

A Python program was developed to decrypt the given substitution cipher. Frequency analysis and substitution mapping were used to recover the original plaintext. The program successfully converted the ciphertext into readable text.

**Figure 3: Substitution Cipher Output:**

```
18 else:
19     plaintext += ch
20
21 print("\nRecovered Plaintext:")
22 print(plaintext)
```

Problems Output Debug Console **Terminal** Ports

PS C:\Users\kaysu\NSC LAB> python substitution\_cipher.py  
Enter Cipher Text: Nb mznv rh Pzrhzm Zirw

Recovered Plaintext:  
My name is Kaisan Arid  
PS C:\Users\kaysu\NSC LAB>

File (NSC LAB) Ln 1, Col 1 Spaces: 4 UTF-8 CRLF {} Python Python 3.14.7

## **4.1 PGP Encryption:**

PGP (Pretty Good Privacy) is a public-key encryption system used for secure file and email communication. In this task, a confidential file was encrypted using the recipient's public key. The encrypted file was then decrypted using the recipient's private key, ensuring that only the intended recipient could access the original data.

### **Import Recipient's Public Key**

First, Shaila imports Sheema's public key into the PGP keyring so that the file can be encrypted for the intended recipient.

### **Encrypt the File**

Shaila selects the confidential file and encrypts it using Sheema's public key. The original file is converted into encrypted ciphertext.

### **Verify Encryption**

After encryption, the encrypted file cannot be read without the corresponding private key. This confirms that the data is securely protected.

### **Send the Encrypted File**

Shaila sends the encrypted file to Sheema via email.

### **Verify Public Key (Optional)**

Before encryption, Shaila verifies Sheema's public key fingerprint to ensure that it belongs to the correct recipient and to prevent man-in-the-middle attacks.

### **Decrypt the File**

Sheema receives the encrypted file and decrypts it using her private key. The original readable file is successfully recovered.

PGP provides secure communication by using public-key cryptography. It ensures confidentiality, authentication, and secure file sharing between users.

File Certificates Tools View Settings Help



Sign/Encrypt



Decrypt/Verify



Import



Export



Certify



Lookup on Server



Notepad



Smartcards



Groups

Enter search term <Alt+Q>

All



All



| Name   | E-Mail           | Status    | Valid From | Valid Until |
|--------|------------------|-----------|------------|-------------|
| shaila | shaila@gmail.com | certified | 8/7/2026   | 8/7/2029    |

Kleopatra

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Import

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Certify

Lookup on Server

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Groups

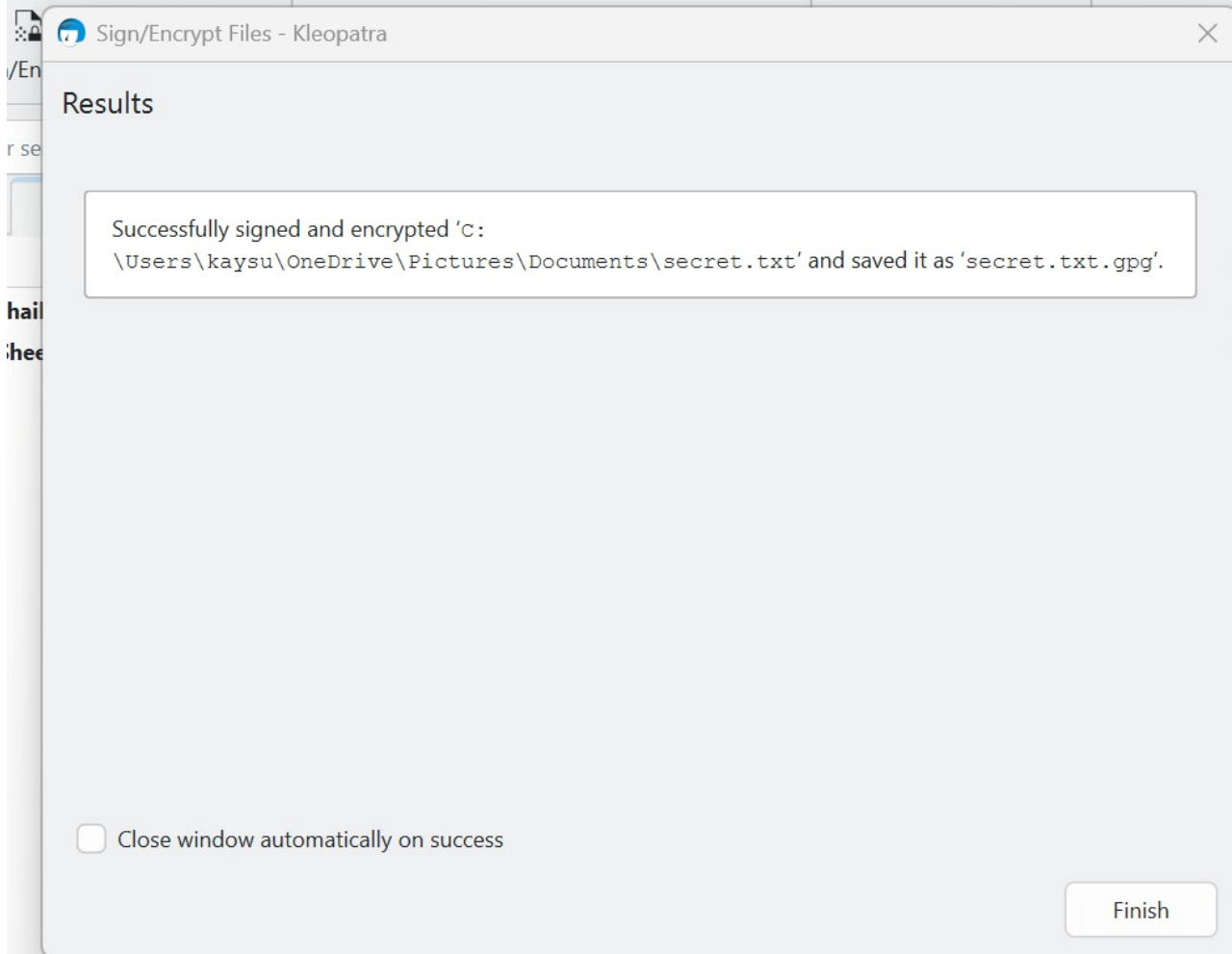
Enter search term <Alt+Q>

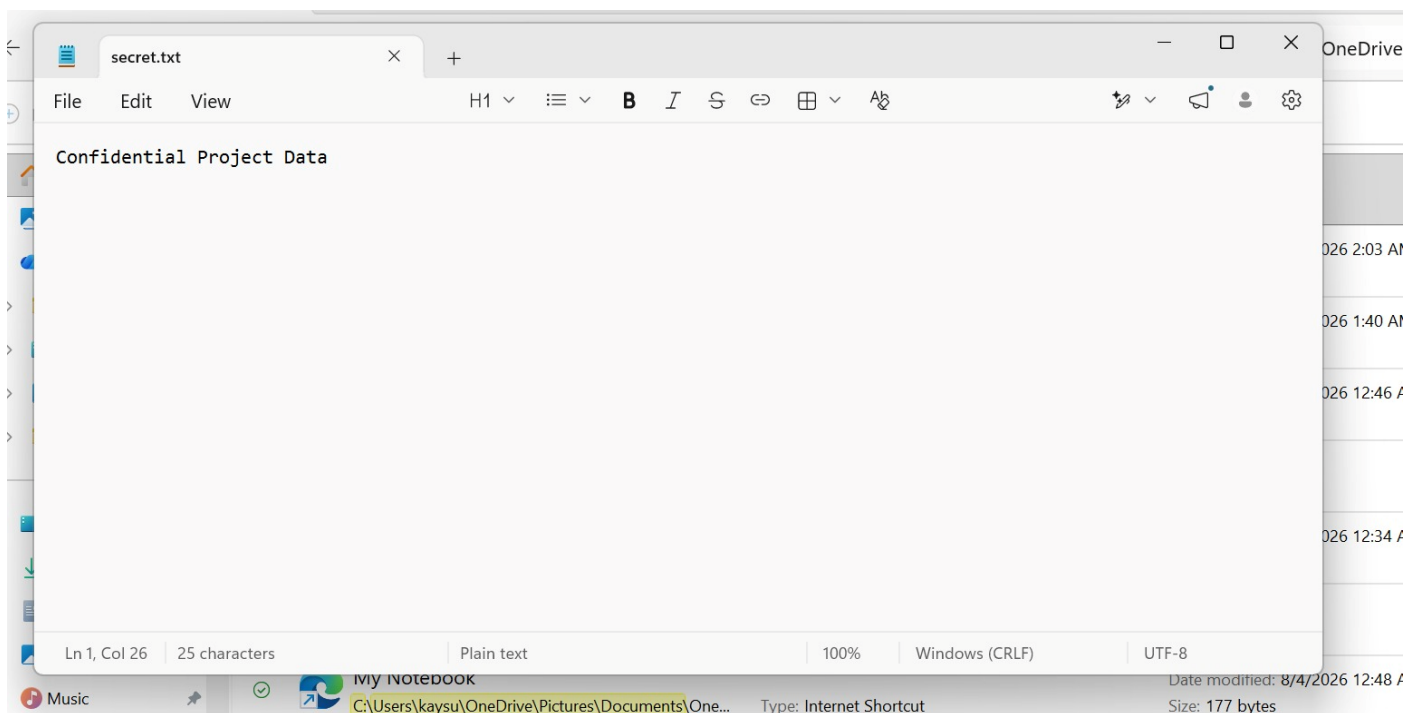
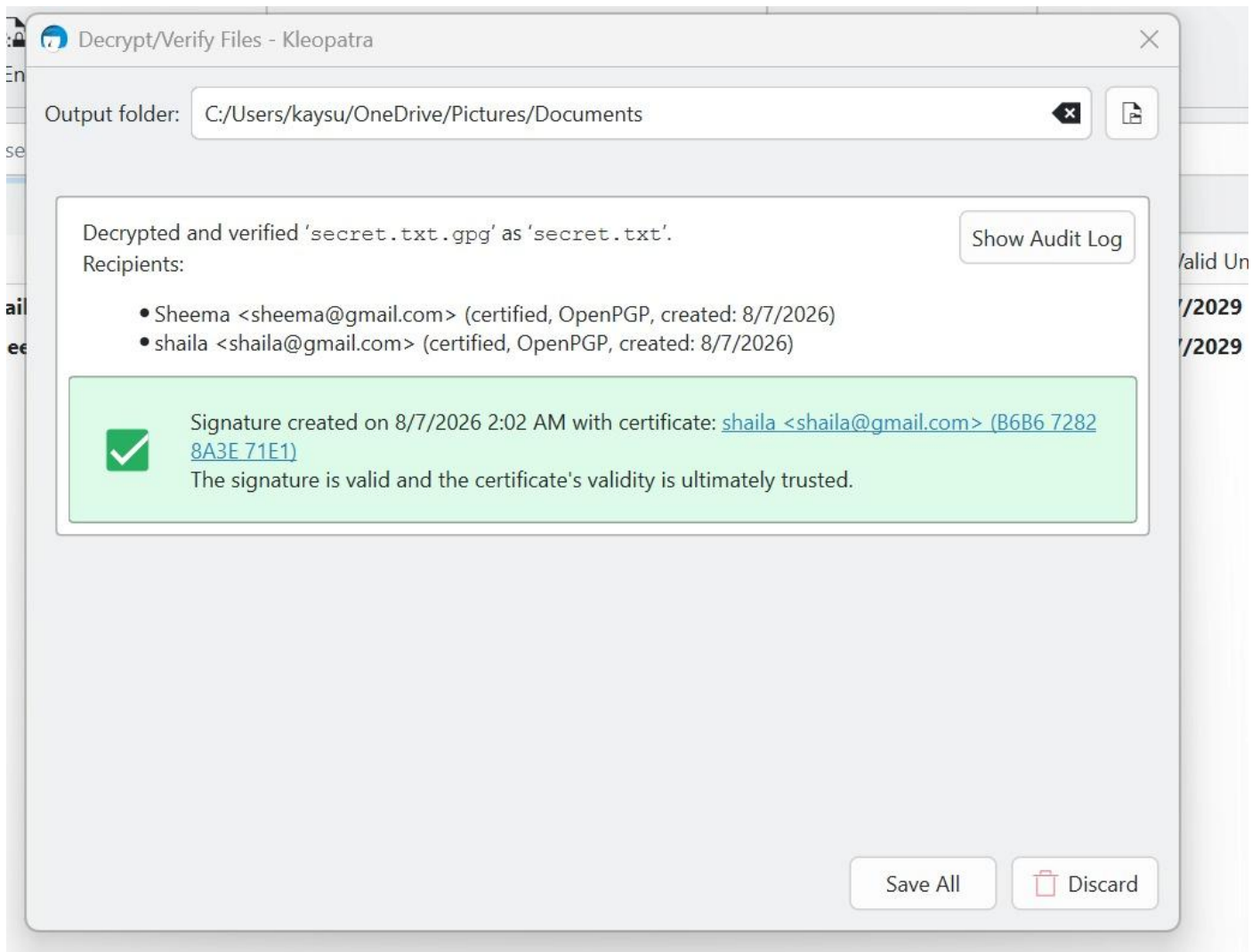
All

All

| Name   | E-Mail           | Status    | Valid From | Valid Until |
|--------|------------------|-----------|------------|-------------|
| shaila | shaila@gmail.com | certified | 8/7/2026   | 8/7/2029    |
| Sheema | sheema@gmail.com | certified | 8/7/2026   | 8/7/2029    |







**Conclusion:**

This lab assignment provided practical experience with several important cryptographic techniques used in network security. AES-128 was implemented for encryption and decryption, while SHA-256 was used to generate and compare hash values and demonstrate the avalanche effect. A substitution-cipher decryption program was developed in Python to recover plaintext from encrypted text. PGP was also demonstrated for securely encrypting and decrypting a confidential file using public-key cryptography. The lab helped demonstrate how encryption, hashing, classical cipher analysis, and public-key cryptography can be used to protect information, maintain data integrity, and support secure communication. The practical implementations also improved understanding of how cryptographic techniques are applied in real-world security scenarios.

**References:**

- Gpg4win, GnuPG for Windows – Official Website. [Gpg4win Official Website](#)
- Python Software Foundation, Python Documentation. [Python Documentation](#)
- MDN Web Docs, Web Crypto API. [Web Crypto API Documentation](#)